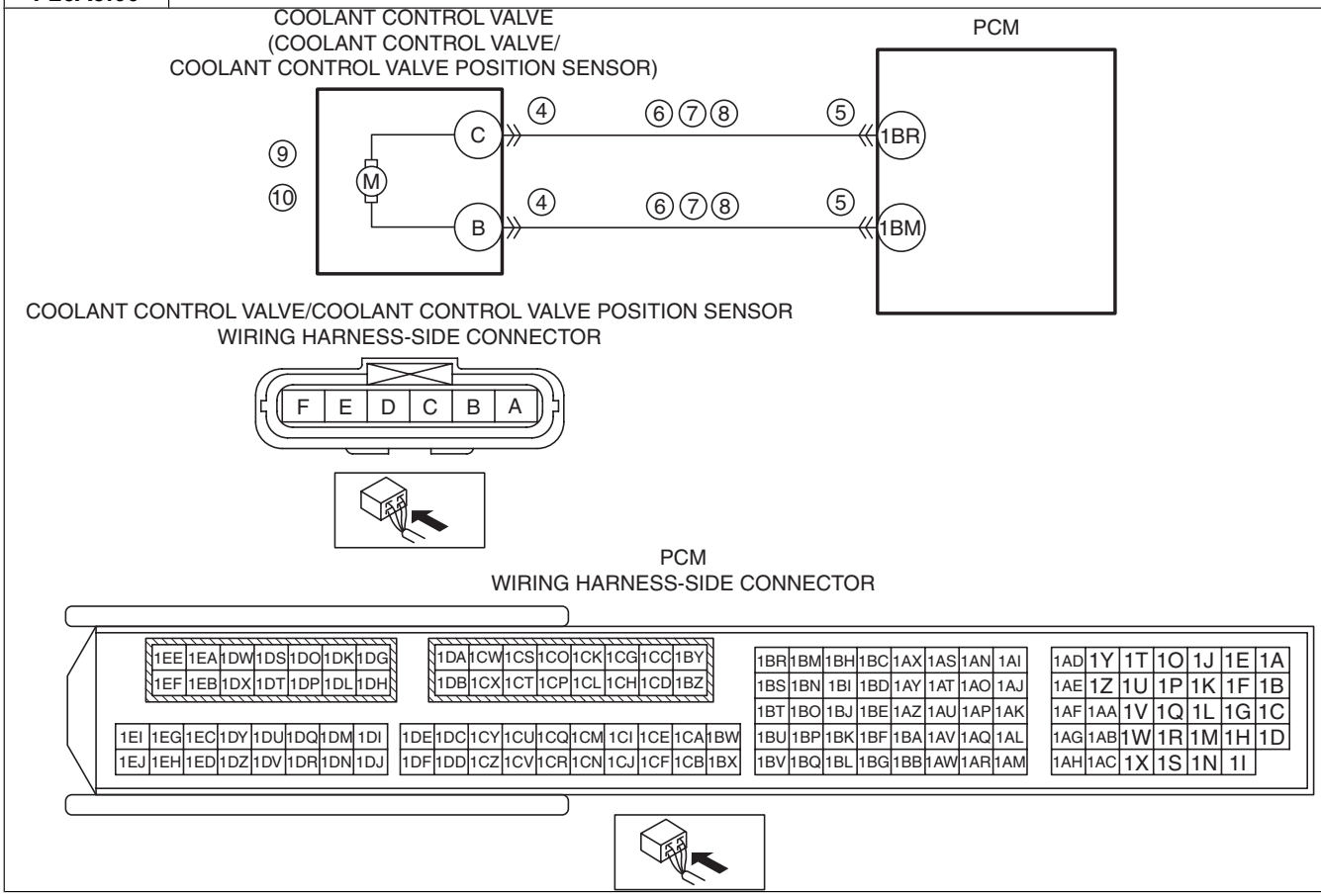
DTC P26A3:00 [SKYACTIV-D 1.5]

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DTC P26A3:00	Coolant control valve control: high duty malfunction
DETECTION CONDITION	• When the following conditions are met, the engine coolant control valve actuator control duty value determined by the engine coolant temperature and intake air temperature exceeds the threshold for a continuous specified time. **MONITORING CONDITIONS** — Engine coolant temperature: **-30—120 °C {-22—248 °F}** — Battery voltage: **8 V or more** — During coolant control valve control — The following DTCs are not detected: • PCM internal processor malfunction: P0606:00 • ECT sensor: P0117:00, P0118:00 • Coolant control valve position sensor: P2682:00, P2683:00 **Diagnostic support note** • This is a continuous monitor (CCM). • The check engine light illuminates if the PCM detects the above malfunction condition during the first drive cycle. • FREEZE FRAME DATA/Snapshot data is available. • DTC is stored in the PCM memory.
FAIL-SAFE FUNCTION	• PCM restricts engine torque. • Inhibits the EGR control. • Inhibits the auto diesel particulate filter regeneration control. • Inhibits the compulsory diesel particulate filter regeneration control. • Inhibits engine-stop by operating the i-stop function.
POSSIBLE CAUSE	• Coolant control valve connector or terminals malfunction • PCM connector or terminals malfunction • Short to ground in wiring harness between the following terminals: — Coolant control valve terminal B—PCM terminal 1BM — Coolant control valve terminal C—PCM terminal 1BR • Short to power supply in wiring harness between the following terminals: — Coolant control valve terminal B—PCM terminal 1BM — Coolant control valve terminal C—PCM terminal 1BR • Open circuit in wiring harness between the following terminals: — Coolant control valve terminal B—PCM terminal 1BM — Coolant control valve terminal C—PCM terminal 1BR • Coolant control valve malfunction • Coolant control valve position sensor malfunction • PCM malfunction

DTC P26A3:00	Coolant control valve control: high duty malfunction



Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**RECORD VEHICLE STATUS WHEN DTC WAS DETECTED TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture of the PC function. • Record the freeze frame data/snap shot data.	—	Go to the next step.
2	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
3	**VERIFY RELATED PENDING CODE AND/OR DTC** • Switch the ignition off, then ON (engine off). • Perform the Pending Trouble Code Access Procedure and DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Are any other PENDING CODEs and/or DTCs present?	Yes	Go to the applicable PENDING CODE or DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	Go to the next step.
4	**INSPECT COOLANT CONTROL VALVE CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the coolant control valve connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 11.
		No	Go to the next step.

STEP	INSPECTION		ACTION
5	**INSPECT PCM CONNECTOR CONDITION** • Disconnect the PCM connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 11.
		No	Go to the next step.
6	**INSPECT COOLANT CONTROL VALVE CIRCUIT FOR SHORT TO GROUND** • Verify that the coolant control valve and PCM connectors are disconnected. • Inspect for continuity between the following terminals (wiring harness-side) and body ground: — Coolant control valve terminal C — Coolant control valve terminal B • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Coolant control valve terminal C—PCM terminal 1BR • Coolant control valve terminal B—PCM terminal 1BM If there is a common connector: • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. If there is no common connector: • Repair or replace the wiring harness which has a short to ground. Go to Step 11.
		No	Go to the next step.
7	**INSPECT COOLANT CONTROL VALVE CIRCUIT FOR SHORT TO POWER SUPPLY** • Verify that the coolant control valve and PCM connectors are disconnected. • Switch the ignition off. **Note** • Another DTC may be stored by the PCM detecting an open circuit. • Measure the voltage between following terminals (wiring harness-side) and ground. — Coolant control valve terminal C — Coolant control valve terminal B • Is the voltage 0 V?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Coolant control valve terminal C—PCM terminal 1BR • Coolant control valve terminal B—PCM terminal 1BM If there is a common connector: • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to power supply. • Repair or replace the malfunctioning part. If there is no common connector: • Repair or replace the wiring harness which has a short to power supply. Go to Step 11.
8	**INSPECT COOLANT CONTROL VALVE CONTROL CIRCUIT FOR OPEN CIRCUIT** • Switch the ignition off. • Verify that the coolant control valve and PCM connectors are disconnected. • Inspect for continuity between the following terminals (wiring harness-side): — Coolant control valve terminal C—PCM terminal 1BR — Coolant control valve terminal B—PCM terminal 1BM • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Coolant control valve terminal C—PCM terminal 1BR • Coolant control valve terminal B—PCM terminal 1BM If there is a common connector: • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. If there is no common connector: • Repair or replace the wiring harness which has an open circuit. Go to Step 11.
9	**INSPECT COOLANT CONTROL VALVE** • Inspect the coolant control valve. (See COOLANT CONTROL VALVE INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the coolant control valve, then go to the next step. (See COOLANT CONTROL VALVE REMOVAL/ INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
10	**INSPECT COOLANT CONTROL VALVE POSITION SENSOR** • Inspect the coolant control valve position sensor. . (See COOLANT CONTROL VALVE POSITION SENSOR INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the coolant control valve position sensor, then go to the next step. (See COOLANT CONTROL VALVE REMOVAL/ INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.

STEP	INSPECTION		ACTION
11	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Perform the KOEO self test. (See KOEO/KOER SELF TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.
12	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.